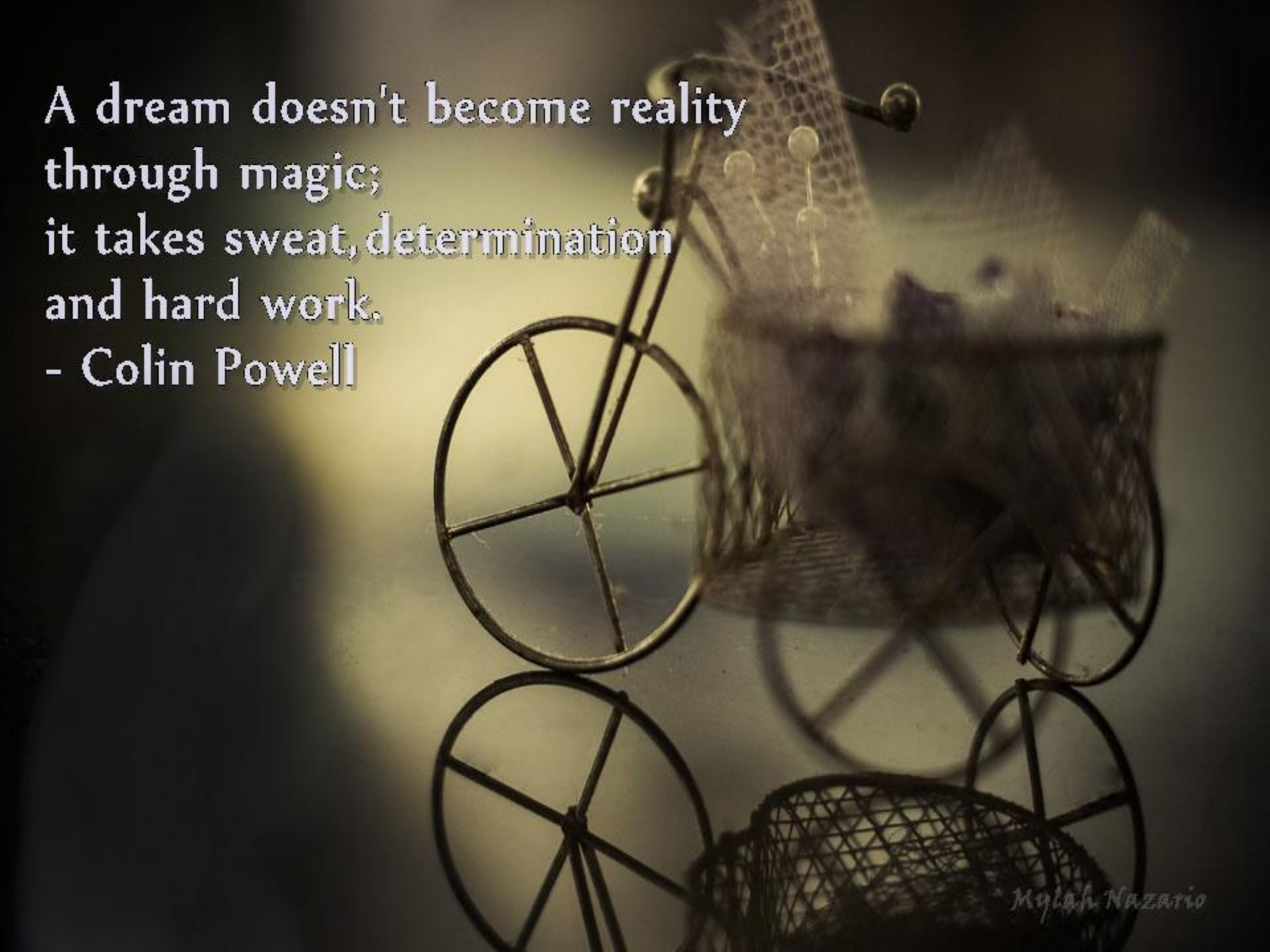

A dream doesn't become reality
through magic;
it takes sweat, determination
and hard work.
- Colin Powell
Mylah Nazario

Example

<u>Inasmuch as</u> all persons have dignity

<u>Since</u> all Filipinos are persons.

<u>It follows that</u> all Filipinos have dignity

Example

all persons have dignity

all Filipinos are persons.

:. all Filipinos have dignity

Example

all persons have dignity (Subject)

all Filipinos are persons. (Predicate)

:. all Filipinos have dignity

M p Figure 1

S m

:. S p

Example

(A) all persons have dignity (Subject)

(A) all Filipinos are persons. (Predicate)

(A):. all Filipinos have dignity

M p Figure 1-bArbArA

S m VALID

:. S p

FIRST FIGURE (SUB-PRE)

Prisoners are law-offenders;

But, some criminals are prisoners;

Therefore, some criminals are law-offenders.

M p

s m

----

s p

FIG 1-DARII
VALID

SECOND FIGURE (PRE-PRE)

A square is a four-sided polygon;

But, a circle is not a four-sided polygon;

A circle is not a square.

P m

S M

S P

FIG 2- CAMESTRES VALID

THIRD FIGURE (SUB-SUB)

Some that glitter are not gold;

But, all that glitter are precious;

Some precious things are not gold.

m P

M s

---

s P

FIG 3- BOCARDO

VALID

FOURTH FIGURE (PRE-SUB)

Some employees are professors;

But, all professors are educators;

Some educators are employees.

p m

M s

---

s p

FIG 4- DIMARIS VALID

CATEGORICAL SYLLOGISMS

Rules and determining fallacies

The Rules of Categorical Syllogism

The rules are divided into two:

Rules 1-4

Rules governing the **terms**

Rules 5-8

Rules governing the **propositions**

RULE 1: The **middle term** must always be taken in the same sense.

FALLACY OF EQUIVOCATION

A small hand-held device dragged to move a cursor on a computer screen is a small rodent that has relatively large ears and eyes and a long tail.





RULE 1: The **middle term** must always be taken in the same sense.

FALLACY OF EQUIVOCATION

1) Jesus is the Word of God.
(2) The Bible is the Word of God.

Therefore:

(3) Jesus is the Bible.

RULE 2: The Major Term and Minor Term cannot have a greater extension in the conclusion than in the premise.

ILLICIT MAJOR

The predicate is **universal in the conclusion**, not in the major premise

ILLICIT MINOR

The subject is **universal in the conclusion**, not in the minor premise

MEMORIZE!

M	Middle term (universal)
P	Major term (universal)
S	Minor term (universal)
m	Niddle term (particular)
p	major term (particular)
s	minor term (particular)
+	affirmative proposition
-	negative proposition

REVIEW: QUANTITY OF SUBJECT AND PREDICATE

S

Look forthe quantifier
Particular (small)
Universal (capital)

P

Look for the quality
A/I (+) = particular (small)
E/O(-) = universal (capital)

EXAMPLE

All engineering students are joining the fun.

EXAMPLE

All engineering students are joining the fun.

S	+	p

EXAMPLE

Some dogs are not healthy.

EXAMPLE

Some dogs are not healthy.

S	**-**	**P**

EXAMPLE

No flowers are in bloom.

EXAMPLE

No flowers are in bloom.

S	-	P

EXAMPLE

Some drinks are potent.

EXAMPLE

Some drinks are potent.

s	+	p

Exercise:

1. Not all foods are expensive.
2. No logicians are irrational.
3. All artists are creative.
4. Some butterflies are big.
5. Some works are not easy.
6. All tables are not yet clean.
7. Some actors are good.

Every student is good; M + p

But, Linda is a student ; S + m

:. Linda is good. S + p

Fig. 1 Barbara

All men are animals; M + p

But, all men are living things; M + s

:. Some living things are animals. s + p

Fig. 3 Darapti

All hammers are tools; M + p

But, no chisels are hammers; S - M

:. No chisels are tools. S - P

FALLACY OF ILLICIT MAJOR TERM



The effect (conclusion) becomes **greater than the cause** (premises), which is **impossible**.

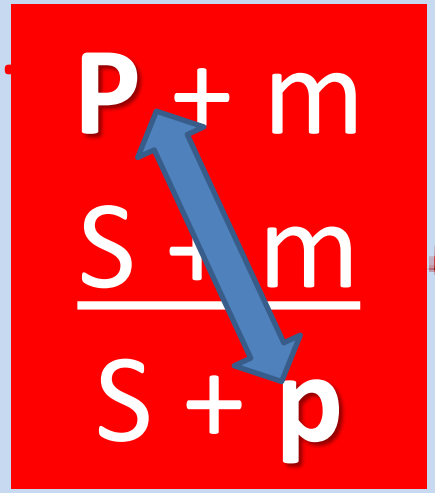
FALLACY OF ILLICIT MAJOR TERM

- **P- major term** should always have smaller or at least the same quantity in the conclusion than in the premises.

Major Premise: All persons have dignity.

Minor Premise: All Filipinos have dignity.

Conclusion: Therefore, all Filipinos are persons.





FALLACY OF ILLICIT MAJOR TERM

- **P- major term** should always have smaller or at least the same quantity in the conclusion than in the premises.

Major Premise: All dogs are **animals.**

Minor Premise: No cats are dogs.

Conclusion: Therefore, no cats are **animals.**

M+ **p**

S - M

S - **P**



FALLACY OF ILLICIT MINOR TERM

- **S- minor term** should always have smaller or at least the same quantity in the conclusion than in the premises.

Major Premise: All persons have dignity.

Minor Premise: All **Filipinos** have dignity.

Conclusion: Therefore, all **Filipinos** are persons.

P + m

S + m

S + p



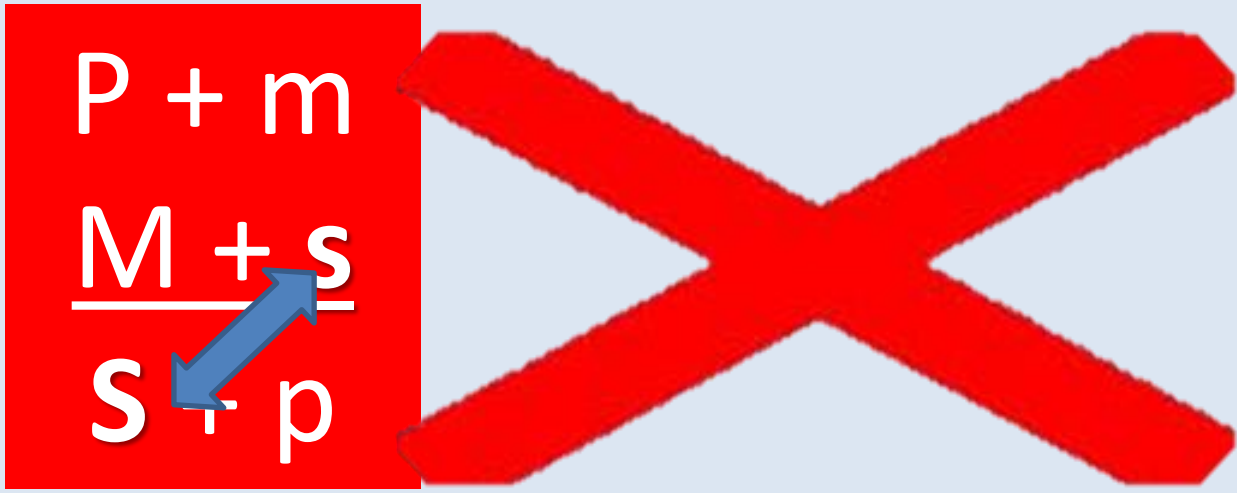
FALLACY OF ILLICIT MINOR TERM

- **S- minor term** should always have smaller or at least the same quantity in the conclusion than in the premises.

Major Premise: All whales are mammals.

Minor Premise: All mammals are **animals**.

Conclusion: Therefore, all **animals** are whales.



RULE 1: The **middle term** must always be taken in the same sense.

FALLACY OF EQUIVOCATION

RULE 2: The Major Term and Minor Term cannot have a greater extension in the conclusion than in the premise.

FALLACY OF ILLICIT MAJOR TERM

FALLACY OF ILLICIT MINOR TERM

RULE 3: The **middle term** should not occur in the conclusion.

FALLACY OF MISPLACED MIDDLE TERM

A steward is a flight attendant; M + p

But, a steward is a male; M + s

:. A steward is a male flight attendant.

M + sp

FALLACY OF MISPLACED MIDDLE TERM

Middle term should NEVER be in the conclusion

Major Premise: All persons have dignity.

Minor Premise: All Filipinos have dignity.

Conclusion: Therefore, all Filipinos are persons.

FALLACY OF MISPLACED MIDDLE TERM

Middle term should NEVER be in the conclusion



M[illegible]r Premise: [illegible] persons have dignity.

Minor P[illegible]e: All Filipinos have dignity.

Con[illegible]ion: Therefore, all Filipinos are people with dignity.

RULE 4: The **middle term** must be distributed universally, at least once, in the premises

FALLACY OF UNDISTRIBUTED MIDDLE TERM



What is true of its whole is true of its parts.

FALLACY OF UNDISTRIBUTED MIDDLE TERM

One or both of the middle terms should be universal.

Major Premise: All persons have dignity.

Minor Premise: All Filipinos have dignity.

Conclusion: Therefore, all Filipinos are persons.

FALLACY OF UNDISTRIBUTED MIDDLE TERM

One or both of the middle terms should be universal.

Major Premise: All persons have dignity.

Minor Premise: All Filipinos have dignity.

Conclusion: Therefore, all Filipinos are persons.

P + m

S + m

S + p



All stewardesses are females. P + m

But, all mothers are females. S + m

:. All mothers are stewardesses. S + p

All apples are fruits P + m

But, some fruits are mangoes m + s

:. Some mangoes are apples s + p

The rules are divided into two:

Rules 1-4

Rules governing the **terms**

Fallacies

1. Equivocation
2. Illicit major term
3. Illicit minor term
4. Misplaced middle term
5. Undistributed Middle Term

The rules are divided into two:

Rules 5-8

Rules governing the **propositions**

Fallacies

1. Negative Conclusion Drawn from Affirmative Premises
2. Exclusive Premises
3. Two Particular Premises
4. Universal Conclusion Drawn froma Particular Premise
5. Affirmative Conclusion Drawn from a Negative Premise

RULE 5: Two affirmative premises **cannot** give a negative conclusion

FALLACY OF A NEGATIVE CONCLUSION DRAWN FROM AFFIRMATIVE PREMISES



A negative conclusion would express **something that is not contained in the premises**.

All stones are hard;
Some diamonds are stone;
:. Some diamonds are not hard.

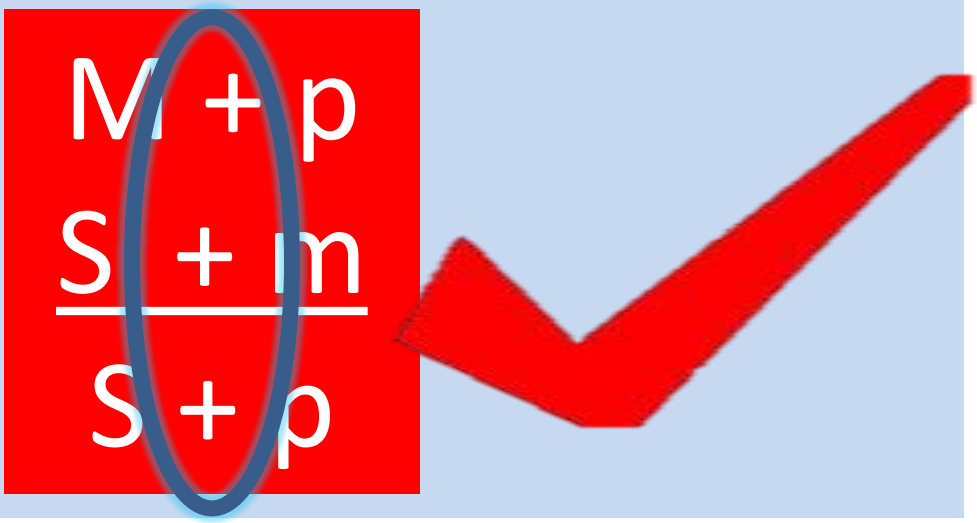
All stones are hard; +

Some diamonds are stone; +

:. Some diamonds are not hard. -

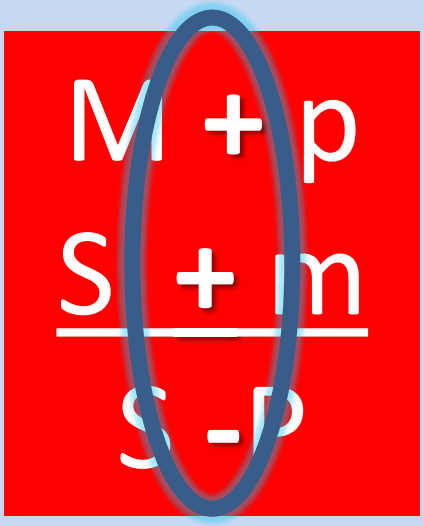
FALLACY OF A NEGATIVE CONCLUSION DRAWN FROM AFFIRMATIVE PREMISES

- Major Premise: All fish have fins.
  Minor Premise: All goldfish are fish.
  Conclusion: Therefore, all goldfish have fins.



FALLACY OF A NEGATIVE CONCLUSION DRAWN FROM AFFIRMATIVE PREMISES

- Major Premise: All fish have fins.
  Minor Premise: All goldfish are fish.
  Conclusion: Therefore, all goldfish do not have fins.





RULE 6: From two negatives, nothing follows.

FALLACY OF EXCLUSIVE/NEGATIVE PREMISES



No relationship can be established ☹

No conclusion can be inferred.

A chair is not a table;

But, a table is not a pen;

:. A pen is not a chair.

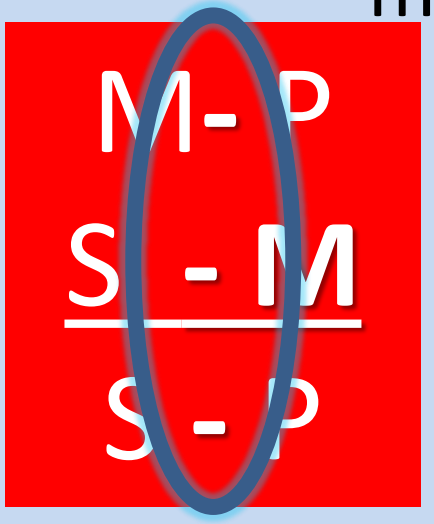
A chair is not a table;

But, a table is not a pen;

:. A pen is not a chair.

FALLACY OF EXCLUSIVE/NEGATIVE PREMISES

- Major Premise: No reptiles are mammals.
- Minor Premise: No dogs are reptiles.
- Conclusion: Therefore, no dogs are mammals.





RULE 7: From two particular premises, nothing follows.

FALLACY OF TWO PARTICULAR PREMISES

a

If both particular premises are **affirmative**, then the subjects and predicates are **particular** (violates rule #4

b

If both particular premises are **negative**, then the syllogism violates rule #6

c

If one particular premise is **affirmative**, and the other **negative**, then the syllogisms are fallacious

Some cats are pets;

But, some pythons are not cats;

:. Some pythons are not pets.

Some cats are pets; m + p

But, some pythons are not cats; s – M

:. Some pythons are not pets. s – P

Some cats are pets; **m + p**

But, some pythons are not cats; **s – M**

:. Some pythons are not pets. **s – P**

Some horseback riders are not males;

But, some drivers are horseback riders;

:. Some drivers are not males.

Some cats are pets;	m + p
But, some pythons are not cats;	s – M
:. Some pythons are not pets.	s – P

Some horseback riders are not males;	m - P
But, some drivers are horseback riders;	s + m
:. Some drivers are not males.	s – P

FALLACY OF TWO PARTICULAR PREMISES

- One or both of the premises should either be E or A

Major Premise: Some reptiles are mammals.

Minor Premise: Some dogs are reptiles.

Conclusion: Therefore, some dogs are mammals.

FALLACY OF TWO PARTICULAR PREMISES

- One or both of the premises should either be E or A

Major Premise: Some reptiles are mammals.

Minor Premise: Some dogs are reptiles.

Conclusion: Therefore, some dogs are mammals.

I m + p
I s + m
I s + p



RULE 8: The conclusion follows the weaker premise.

FALLACY OF THE UNIVERSAL CONCLUSION DRAWN FROM A PARTICULAR PREMISE



If one premise is **universal**, and the other is **particular**, the conclusion should be **particular**

All mongoloids are mentally retarded;
But, some mongoloids are special children;
:. All special children are mentally retarded.

All mongoloids are mentally retarded; M + p

But, some mongoloids are special children; m + s

:. All special children are mentally retarded. S + p

FALLACY OF A UNIVERSAL CONCLUSION DRAWN FROM A PARTICULAR PREMISE

Major premise: All cats are marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, all mammals are marsupials.

FALLACY OF A UNIVERSAL CONCLUSION DRAWN FROM A PARTICULAR PREMISE

Major premise: All cats are marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, all mammals are marsupials.

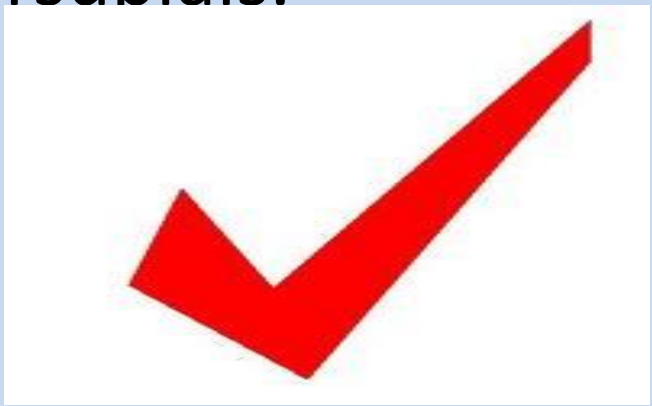
A	M + p
I	s + m
A	S + p



FALLACY OF A UNIVERSAL CONCLUSION DRAWN FROM A PARTICULAR PREMISE

Major premise: All cats are marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, some mammals are marsupials.

A	M + p
I	s + m
I	s + p

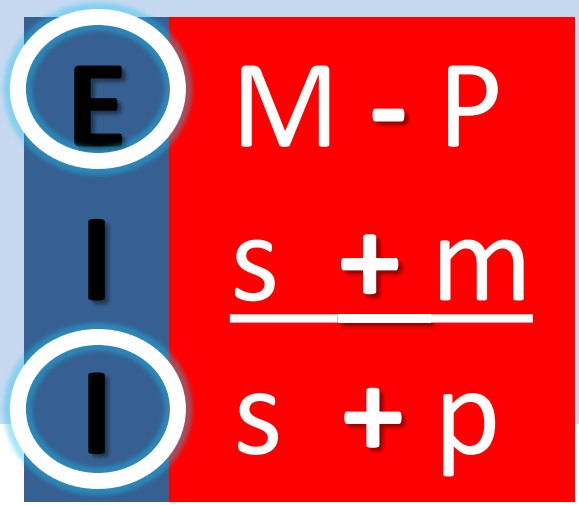


FALLACY OF A AFFIRMATIVE CONCLUSION DRAWN FROM A NEGATIVE PREMISE

Major premise: All cats are NOT marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, some mammals are marsupials.

FALLACY OF A AFFIRMATIVE CONCLUSION DRAWN FROM A NEGATIVE PREMISE

Major premise: All cats are NOT marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, some mammals are marsupials.





FALLACY OF A AFFIRMATIVE CONCLUSION DRAWN FROM A NEGATIVE PREMISE

Major premise: All cats are NOT marsupials.
Minor Premise: Some mammals are cats.
Conclusion: Therefore, some mammals are not marsupials.

E	M - P
I	s + m
O	s - P



DRILLS

Identify the fallacy(ies) committed.

Rape is immoral;
But, cheating is immoral;
:. Cheating is rape.

A martyr dies for a noble cause;

But, some wives are martyrs;

:. Some wives do not die for a noble cause.

Waiters wait;

But, true love waits;

:. True love is a waiter.

Horror stories send chilling effects;

But, murder stories are horror stories;

:. Murder stories don’t send chilling effects.

CATEGORICAL SYLLOGISMS

Rules and determining fallacies